

Beckham Moreland

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Education

San Diego State University, College of Engineering
B.S. in Aerospace Engineering

Expected May 2029
Cumulative GPA: 3.7/4.0

Experience

UAS Engineering Intern, NASA Ames (AMA) – Moffett Field, CA Jun 2026 – Present

- Built and iterated 3 helicopter-deployable SAR multirotors in 3 weeks, translating 5 Air National Guard consultations and a 6-concept trade study into vehicle requirements; first prototype flight-ready in 4 days
- Designed motor mounts, body, landing fins, internal structures, and avionics packaging in Fusion 360; fabricated/integrated flight hardware and transitioned from PLA to PETG after test-identified damage
- Reduced component mass 20%+ through FEA-driven redesign under maximum motor-thrust and downwash loads, revising motor-plate/internal structures while maintaining analyzed structural margins
- Designed and executed 15+ wind-tunnel runs varying CG, yaw, and 10–85 ft/s flow under forward-flight and downwash-like conditions; analyzed six-axis load data/PX4 logs and authored TRR material to validate redesign stability gains
- Designed a removable CG ballast fixture and executed 10+ flight/CG tests plus three 30-ft carrier-UAS releases; used results to drive mass-distribution, landing-fin, and airframe changes

Test and Visualization Engineering Intern, MeshMap Labs Inc. – San Francisco, CA May 2025 – Sep 2025

- Led UAV and RTK field-test campaigns across 4 sites, varying altitude, image overlap, and capture parameters to establish repeatable data-collection procedures under changing operating conditions
- Diagnosed coverage and data-quality failures through post-flight analysis, planned corrective missions, and verified revised workflows in georeferenced models combining UAV imagery and terrestrial LiDAR

Full Stack Developer Intern, Axyz Technologies – Oakland, CA Jun 2023 – Jul 2025

- Built and shipped 12+ Python/Django and JavaScript features from user requirements, debugging application/database issues and iterating through testing and stakeholder feedback

Skills

Mechanical Design & Analysis: SolidWorks, Fusion 360, CAD assemblies/drawings, structural FEA, design trade studies, lightweight design

Prototyping & Integration: Additive manufacturing, threaded inserts, electromechanical assembly, avionics packaging, motors/ESCs, sensors/actuators, soldering/wiring

Test & Engineering: Test planning/matrices, test fixtures, test readiness reviews, wind-tunnel/flight testing, six-axis load-cell DAQ, root-cause investigation, Python, MATLAB

Projects

Fault-Tolerant Navigation Bench, Personal Project In Progress

- Designed a bench-scale GNSS/IMU validation platform, defining hardware architecture, MCU/sensor interfaces, logging, and test requirements for repeatable hardware verification
- Developing Python fault-injection tools for GPS loss, IMU bias/noise, frozen telemetry, and signal dropouts to characterize detection and recovery behavior before UAV integration

Autonomous UAV Systems Research, SDSU Undergraduate Research Mar 2025 – Present

- Integrating onboard sensors, telemetry links, and ground-station controls for an experimental UAV; documenting interfaces and validating communications ahead of hardware and flight testing
- Evaluating MATLAB controls and state-estimation outputs through error analysis and test documentation, identifying validation needs before implementation on the physical aircraft